Science Inquiry

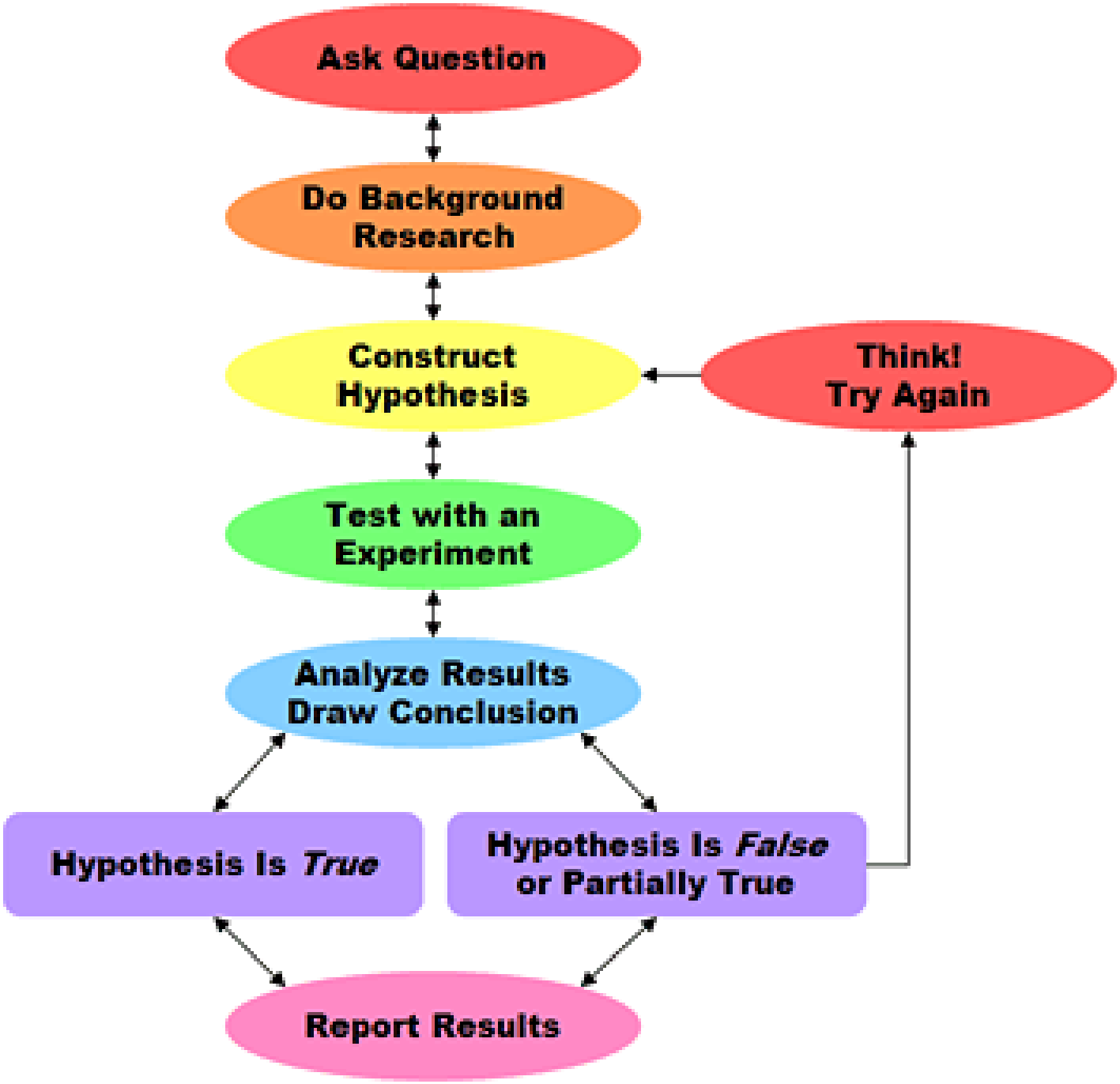
Scientific Method

Planning an Investigation

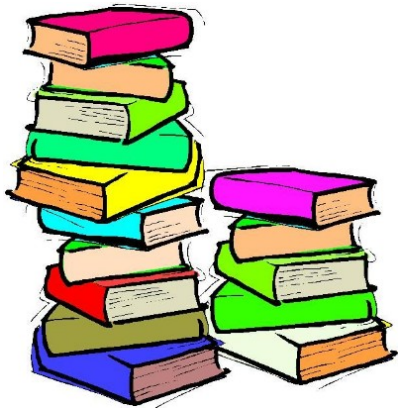
Scientific Method

- A way to investigate, that allows accurate assessment of whether a hypothesis is supported by the data, incorrect or inconclusive.

- Each experiment/investigation adds a small piece to a much larger body of knowledge.


Ask Question
Do Background Research
Construct Hypothesis
Think! Try Again
Test with an Experiment
Analyze Results Draw Conclusion
Hypothesis Is *True*
Hypothesis Is *False* or Partially True
Report Results

Literature Review



- Reading previous research about the subject.
  - Helps to define the problem/question
  - Prevents unnecessary duplication – can build on prior knowledge.
  - To assess research methods used, and learn from these
  - Helps scientists to related findings to prior body of knowledge
  - Allows consideration of areas for further research

Safety

- Important to consider safety of research team and participants.
  - Protective clothing
  - Ensuring participants feel safe and comfortable
  - Access to medical care/first aid if needed



Ethics

- A set of moral principles. For experiments involving people:
  - Voluntary participation
  - Informed consent
  - No risk of harm
  - Confidentiality



- For animal research:
  - Valid
  - Humane
  - Justifiable
  - Considerate

Selecting Variables

- Should be linked to hypothesis being tested.
- **Independent Variable:** the thing being changed.
- **Dependent Variable:** the thing that changes in response.
- Should be:
  - Measurable
  - Clear
  - Other variables should be controlled (kept the same)
- **Control Sample:** a sample where the independent variable is not changed. Used as a comparison

Experimental Validity

- **Validity:**
  - how well the experiment tests what it is supposed to test
  - whether the results are accurate
  - How confident you can be about the conclusions drawn.

- **Valid experiments:**
  - Have all variables well controlled, except for the IV and DV being tested.
  - Have a neutral control sample to compare.
  - Have variables that are measurable/quantifiable

This allows us to be confident that any change to the DV must be due to the IV, and not another factor.



- **Valid experiments:**
  - Are RELIABLE.
  - Reliability:
    - confidence that the results will show the same thing each time the experiment is run.
    - Helped by large sample sizes, repeated trials.
    - Can help overcome problems with uncontrolled variables/outliers in the results.



- **Valid experiments:**
  - Minimise error
  - Error: mistakes that affect results:

  Human Error: *data entry mistakes, misreading equipment etc.*

  Systematic Error: *mistakes in experimental design*

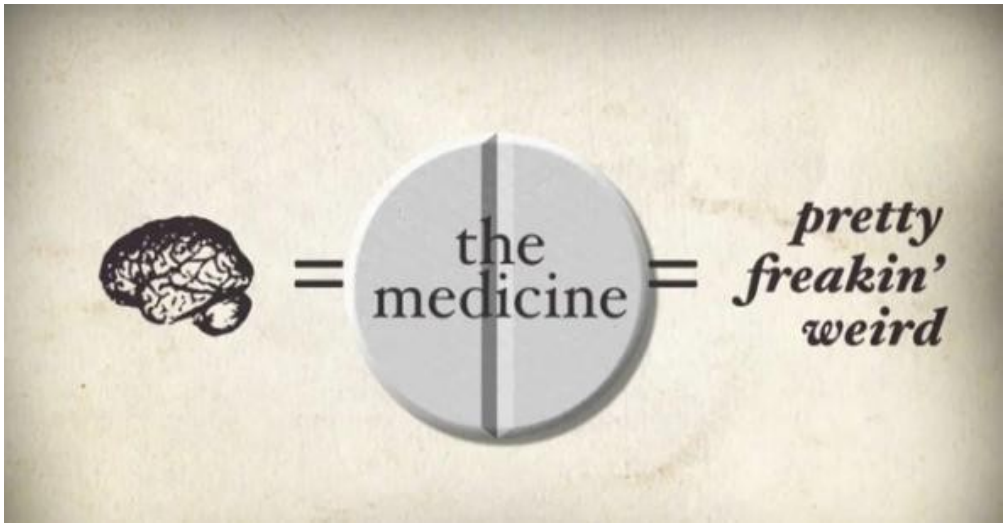
  Random Error: *unpredictable eg environmental conditions*

- **Valid experiments:**
  - Avoid bias:
- Scientists try to be ***objective*.**
  - Try not to let their own thoughts, feelings and opinions to influence their experiment
  - Keep an open mind about what the results may be
  - Try not to have *conflict of interest* eg: accepting money or favours for the result to be a certain way.
  - Allow other scientists to review their experimental design, results and conclusions.

The placebo effect

- Is a type of bias that can occur
- “trick of the mind”
- Scientists or experiment participants may unconsciously introduce bias into the results.

  - Eg: people receiving a medication may *believe* they will get better, and therefore report feeling better even if the symptoms don’t go away.



Avoiding placebo effect

Blind studies:

- Neither the test group nor the controls know whether they are getting the drug or the placebo.

Double Blind studies:

- None of the people being tested, the control group, OR the scientist know which group is getting the drug and which group is getting the placebo until after the experiment.